Jimin Han

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Objective

Undergraduate mechanical engineering student seeking opportunities to apply CAD, prototyping, and computational modeling skills to interdisciplinary research and engineering projects. I aim to deepen my experience with simulation and experimentation while contributing to solutions that improve real-world systems and user experiences.

Education

Rice University, Texas, USA — Aug 2025 – Present
B.S. in Mechanical Engineering and B.A. in Mathematics (expected May 2029), GPA: 4.0/4.0

Skills

Software: MATLAB, Python, LabVIEW; **Design & Analysis**: SolidWorks, Onshape, DIC;
Manufacturing: 3D Printing, Waterjetting, Laser Cutting, Machining, Soldering;
Languages: Korean Native, English Bilingual, Mandarin Working Proficiency

Work Experience

Tribomechadynamics Lab – *Undergraduate Research Assistant* – Rice University — Oct 2025 – Present
Research Advisor: Prof. Matthew Brake

- Design and test a Mini Split Hopkinson Pressure Bar to study high-strain-rate failure in jointed metallic microstructures
- Integrate DIC, high-speed cameras, strain gauges, and LabVIEW workflows to capture real-time stress–strain dynamics
- Develop SolidWorks CAD models, assembly drawings, and BOM documentation for test fixtures, pressure bar components, and safety enclosures

NSF RIDE REU – *Research Intern* – University of Massachusetts Amherst — Jun 2026 – Aug 2026
Research Advisor: Prof. Eric Gonzales

- Selected as 1 of 10 interns for an NSF-funded transportation research program; conducted research on electric-bus operations and mobility efficiency in partnership with BasiGO
- Developed Python pipelines to process 57,000+ Rwanda e-bus telematics records, integrating CCTV passenger counts, GPS elevation, and weather data to engineer features and evaluate machine learning models for electric-bus power prediction
- Presented findings through a research poster and co-authored a manuscript for submission to the Transportation Research Board (TRB)

NSF ExLENT – *Research Trainee* – Michigan Technological University — Feb 2026 – May 2026

- Selected as 1 of 20 fellows for an NSF-funded engineering program focused on mechatronics workforce development
- Completed 40+ hours of training and lab work across electrical systems, hydraulics, PLCs, robotics, AI, and cybersecurity
- Applied mechatronics concepts through FANUC robotics, PLC programming, industrial control systems, AI-based inspection, and industry site visits

Lukkey AG – *Engineering Support Intern* – Remote — Dec 2025 – Jan 2026

- Structured hardware wallet technical specifications for Korean-market localization of a Swiss digital asset product
- Performed data-driven analysis of the Korean crypto hardware market to support product deployment
- Hosted 10 meetings with the engineering team to translate technical terminology and ensure accurate metric conversions

Xcellent Life – *Strategy Intern* – Nashville, TN — Jun 2025 – Jul 2025

- Analyzed regulatory, cultural, and logistical constraints across 3 healthcare-adjacent markets to guide market entry
- Collaborated with the CEO to refine go-to-market messaging and execute a digital awareness campaign
- Built standardized email automation templates, reducing manual outreach time and supporting investor outreach to 10+ contacts, contributing to a multi-investor meeting

Projects

Battle Bots – *Lead & Developer at Rice Robotics Club* – Rice University — Sep 2025 – Present

- Lead two 1.5-hour meetings per week focused on design, fabrication, electronics integration, and competition preparation
- Design and build 5+ plastic antweight bots using Onshape, 3D printing, waterjetting, machining, soldering, and electronics
- Organize internal competitions and coordinate Texas event participation; placed second in the most recent competition

Rice Eclipse – *Recovery Team Engineer* – Rice University — Sep 2025 – Feb 2026

- Collaborated on dual-stage test rockets exceeding Mach 2, focusing on safe and reliable recovery
- Designed recovery modules for airframe separation, pressurization, parachute deployment, and avionics mounting
- Applied OpenRocket, SolidWorks, MATLAB, and fabrication tools to iterate designs and produce launch-ready hardware